实验环境：

服务器	阿里云 ECS (ecs.gn6i-c4g1.xlarge)
GPU	NVIDIA T4 (16GB 显存)
模型	GPT2-XL (1.5B 参数)
框架	EasyEdit

Task 1: 基础环境搭建与基线测试 (Baseline Evaluation)

实验目的：

在不进行任何编辑操作的情况下，测试 GPT2-XL 模型对 10 条特定事实的回答，验证模型在编辑前是否存在知识盲区或过时知识。

实验截图：

```
Task 1: 基线测试 - 编辑前模型回答
============================================================

1. The current CEO of Twitter (X) is
   模型回答: Jack Dorsey, who was hired in 2013.
   期望(旧知识): Elon Musk ❌

2. The capital of Australia is
   模型回答: Canberra, the capital of Australia is Canberra, the
   期望(旧知识): Canberra ✅

3. The founder of Microsoft is
   模型回答: a billionaire, and he's also a philanthropist
   期望(旧知识): Bill Gates ❌

4. The highest mountain in the world is
   模型回答: Mount Everest, and the highest peak in the United
   期望(旧知识): Mount Everest ✅

5. The author of Harry Potter is
   模型回答: a wizard.

The author of Harry Potter
   期望(旧知识): J.K. Rowling ❌

6. The planet known as the Red Planet is
   模型回答: a rocky world with a thick atmosphere and a thick
   期望(旧知识): Mars ❌
```

实验结果：

基线测试结果：4/10 条包含旧知识（40.0%）

统计项	数值
正确回答	4 条
错误回答	6 条
准确率	40.0%

Task 2: 单条事实编辑实践 (Single Fact Editing using ROME

实验目的：

使用 ROME（Rank-One Model Editing）算法对 GPT2-XL 模型中的 10 条事实进行单条编辑，验证该算法能否通过修改前馈神经网络（FFN）特定层权重来精确更新模型知识。

实验设置：

参数	设置
编辑层数	layer 17
优化目标层	layer 47
学习率	0.5
迭代次数	20
精度	float16（半精度）

实验截图：

```
========== 编辑第 1 条 ==========
Prompt: The current CEO of Twitter is...
原事实: Elon Musk → 新事实: Linda Yaccarino
2026-05-13 10:09:49,150 - easyeditor.editors.editor - INFO - Instantiating model
05/13/2026 10:09:49 - INFO - easyeditor.editors.editor -   Instantiating model
Executing ROME algorithm for the update: [The current CEO of Twitter is] -> [ Linda Yaccarino]
Cached context templates ['{}', 'The "Crisis". {}', 'The first thing I did. {}', "Therefore, it's a. {}", 'Therefore, th
e only thing. {}', 'Because the game is not. {}', 'Because of its proximity to. {}', "I'm a bit surprised. {}", "I'm not
 going to. {}", "You can't get a. {}", "You'll need an HTML. {}", 'The "B" in "B.A.. {}', 'The "I\'m a Man" campaign, wh
ich. {}', 'Therefore, we can conclude that there is not only. {}', 'Therefore, the best way to prevent a future catastro
phe. {}', 'Because we are the ones who are being targeted,. {}', 'Because of the recent changes in our policy on the. {}
', "I think that the way we've done it,. {}", 'I\'m a little nervous about this," says one. {}', 'You are the only one w
ho can stop the war. {}', "You're not going to be able to do this. {}"]
Computing left vector (u)...
Selected u projection object Twitter
Left vector shape: torch.Size([6400])
Computing right vector (v)
Lookup index found: 4 | Sentence: The current CEO of Twitter is Linda Yaccar | Token:  Twitter
Rewrite layer is 17
Tying optimization objective to 47
Recording initial value of v*
loss 3.206 = 3.206 + 0.0 + 0.0 avg prob of [ Linda Yaccarino] 0.04097975790500641
loss 2.491 = 2.442 + 0.002 + 0.047 avg prob of [ Linda Yaccarino] 0.08768204599618912
loss 1.981 = 1.89 + 0.006 + 0.085 avg prob of [ Linda Yaccarino] 0.15239204466342926
loss 1.618 = 1.491 + 0.009 + 0.118 avg prob of [ Linda Yaccarino] 0.22756251692771912
loss 1.296 = 1.149 + 0.01 + 0.137 avg prob of [ Linda Yaccarino] 0.31956079602241516
loss 0.994 = 0.848 + 0.008 + 0.137 avg prob of [ Linda Yaccarino] 0.430618554353714
loss 0.773 = 0.627 + 0.008 + 0.137 avg prob of [ Linda Yaccarino] 0.5364371538162231
loss 0.619 = 0.473 + 0.009 + 0.137 avg prob of [ Linda Yaccarino] 0.6250548958778381
```

实验结果：

以 Twitter CEO 编辑为例：

阶段	输入	模型回答
编辑前	The current CEO of Twitter is	Jack Dorsey, who was hired in 2013.
编辑后	The current CEO of Twitter is	Linda Yaccarino, who was previously...

统计项	数值
编辑总数	10 条
成功数	10 条
成功率	**100%**

Task 3: 批量知识编辑实践 (Batch Editing using MEMIT)

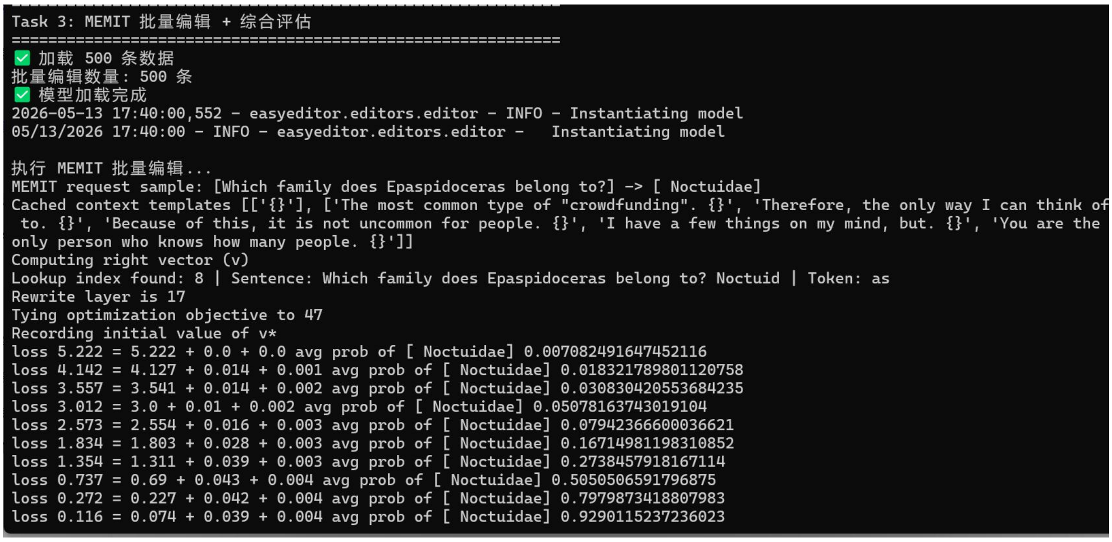
实验目的：

使用 MEMIT（Mass-Editing Memory in a Transformer）算法对 GPT2-XL 模型进行批量知识编辑，一次性向模型中注入 500 条知识，验证该算法在大规模知识更新场景下的性能和效果，并评估其编辑成功率、泛化性及局部性。

实验设置：

参数	设置
数据集	ZsRE（500 条）
编辑层数	layers 13, 14, 15, 16, 17
精度	float16（半精度）
评估样本数	100 条（抽样评估）

实验截图：



```
Task 3: MEMIT 批量编辑 + 综合评估
============================================================
✅ 加载 500 条数据
批量编辑数量: 500 条
✅ 模型加载完成
2026-05-13 17:40:00,552 - easyeditor.editors.editor - INFO - Instantiating model
05/13/2026 17:40:00 - INFO - easyeditor.editors.editor -   Instantiating model

执行 MEMIT 批量编辑...
MEMIT request sample: [Which family does Epaspidoceras belong to?] -> [ Noctuidae]
Cached context templates [['{}'], ['The most common type of "crowdfunding". {}', 'Therefore, the only way I can think of
 to. {}', 'Because of this, it is not uncommon for people. {}', 'I have a few things on my mind, but. {}', 'You are the
only person who knows how many people. {}']]
Computing right vector (v)
Lookup index found: 8 | Sentence: Which family does Epaspidoceras belong to? Noctuid | Token: as
Rewrite layer is 17
Tying optimization objective to 47
Recording initial value of v*
loss 5.222 = 5.222 + 0.0 + 0.0 avg prob of [ Noctuidae] 0.007082491647452116
loss 4.142 = 4.127 + 0.014 + 0.001 avg prob of [ Noctuidae] 0.018321789801120758
loss 3.557 = 3.541 + 0.014 + 0.002 avg prob of [ Noctuidae] 0.030830420553684235
loss 3.012 = 3.0 + 0.01 + 0.002 avg prob of [ Noctuidae] 0.05078163743019104
loss 2.573 = 2.554 + 0.016 + 0.003 avg prob of [ Noctuidae] 0.07942366600036621
loss 1.834 = 1.803 + 0.028 + 0.003 avg prob of [ Noctuidae] 0.16714981198310852
loss 1.354 = 1.311 + 0.039 + 0.003 avg prob of [ Noctuidae] 0.2738457918167114
loss 0.737 = 0.69 + 0.043 + 0.004 avg prob of [ Noctuidae] 0.5050506591796875
loss 0.272 = 0.227 + 0.042 + 0.004 avg prob of [ Noctuidae] 0.7979873418807983
loss 0.116 = 0.074 + 0.039 + 0.004 avg prob of [ Noctuidae] 0.9290115237236023
```

实验结果：

资源消耗记录：

峰值显存占用：10601 MB（约 10.4 GB）

平均每条耗时：8.57 秒/条

实验结果：

指标	得分	说明
ES（编辑成功率）	84.0%	模型对直接编辑的 Prompt 输出目标答案
PS（泛化性）	57.0%	模型对同义改写 Prompt 输出目标答案
NS（局部性）	65.0%	模型对无关事实的回答未受影响

成功案例：

阶段	输入	模型回答
编辑前	Which family does Epaspidoceras belong to?	Aspidoceratidae
编辑后	Which family does Epaspidoceras belong to?	Noctuidae
泛化测试	What family are Epaspidoceras?	Noctuidae

失败案例：

项目	内容
Prompt	Who is listed as Wang Jipeng father?
目标答案	Wang Chonghua
模型实际回答	Wang Yanjun.
原因分析	模型未学习到目标知识，可能因为该条数据在批量编辑中被遗漏或权重更新不足。该实体在同一批次中出现了多次编辑请求，导致参数更新相互覆盖，最终编辑未生效。

Task 4: 综合评估 (Comprehensive Evaluation)

实验设置：

基于 Task 2 (ROME) 的编辑结果进行评估

实验截图：

```
--- 第 1 条: The current CEO of Twitter is... ---
2026-05-13 10:46:42,688 - easyeditor.editors.editor - INFO - Instantiating model
05/13/2026 10:46:42 - INFO - easyeditor.editors.editor -   Instantiating model
Executing ROME algorithm for the update: [The current CEO of Twitter is] -> [ Linda Yaccarino]
Cached context templates ['{}', 'The "B" in. {}', 'The last time we had. {}', 'Therefore, we have an. {}', 'Therefore, I
 would like. {}', "Because I'm a little. {}", 'Because the first thing that. {}', "I'm sorry, you. {}", "I think it's go
ing. {}", 'You can use the following. {}', 'You can find the full. {}', 'The first time I saw it was when it was. {}', '
The last time we had a real debate in the. {}', 'Therefore, the best way to prevent a child from. {}', 'Therefore, if we
 are to be consistent, it. {}', 'Because I am in my 40s, and my. {}', 'Because the game is still in its infancy, there.
{}', 'I have a very strong sense of the need to. {}', "I'm going to be the first one to go. {}", 'You are the only one w
ho can stop him!. {}', 'You are a man of great power and influence.. {}']
Computing left vector (u)...
Selected u projection object Twitter
Left vector shape: torch.Size([6400])
Computing right vector (v)
Lookup index found: 4 | Sentence: The current CEO of Twitter is Linda Yaccar | Token:  Twitter
Rewrite layer is 17
Tying optimization objective to 47
Recording initial value of v*
loss 3.2 = 3.2 + 0.0 + 0.0 avg prob of [ Linda Yaccarino] 0.04113104194402695
loss 2.492 = 2.443 + 0.002 + 0.047 avg prob of [ Linda Yaccarino] 0.08734176307916641
loss 1.99 = 1.899 + 0.006 + 0.085 avg prob of [ Linda Yaccarino] 0.15032489597797394
loss 1.617 = 1.49 + 0.009 + 0.119 avg prob of [ Linda Yaccarino] 0.22742509841918945
loss 1.284 = 1.137 + 0.009 + 0.137 avg prob of [ Linda Yaccarino] 0.3236261010169983
loss 0.982 = 0.837 + 0.008 + 0.137 avg prob of [ Linda Yaccarino] 0.43471473455429077
loss 0.753 = 0.608 + 0.008 + 0.137 avg prob of [ Linda Yaccarino] 0.5451506972312927
loss 0.598 = 0.452 + 0.008 + 0.137 avg prob of [ Linda Yaccarino] 0.637094259262085
loss 0.47 = 0.324 + 0.009 + 0.137 avg prob of [ Linda Yaccarino] 0.7244775295257568
```

```
loss 0.156 = 0.012 + 0.007 + 0.137 avg prob of [ Linda Yaccarino] 0.9884945154190063
loss 0.153 = 0.01 + 0.006 + 0.137 avg prob of [ Linda Yaccarino] 0.9903872609138489
Delta norm: 58.236724853515625
Change in target norm: 14.559181213378906 to 59.61819076538086 => 45.05900955200195
Division Factor: 11.803506851196289
Right vector norm: 4.933849811553955
Right vector shape: torch.Size([1600])
Deltas successfully computed for ['transformer.h.17.mlp.c_proj.weight']
New weights successfully inserted into ['transformer.h.17.mlp.c_proj.weight']
2026-05-13 10:47:15,125 - easyeditor.editors.editor - INFO - Execution 0 editing took 14.627016544342041
05/13/2026 10:47:15 - INFO - easyeditor.editors.editor -   Execution 0 editing took 14.627016544342041
2026-05-13 10:47:15,174 - easyeditor.editors.editor - INFO - Evaluation took 0.048949480056762695
05/13/2026 10:47:15 - INFO - easyeditor.editors.editor -   Evaluation took 0.048949480056762695
2026-05-13 10:47:15,175 - easyeditor.editors.editor - INFO - 0 editing: The current CEO of Twitter is -> Linda Yaccarino

 {'pre': {'rewrite_acc': [0.6], 'portability': {}}, 'case_id': 0, 'requested_rewrite': {'prompt': 'The current CEO of Tw
itter is', 'target_new': 'Linda Yaccarino', 'ground_truth': 'Elon Musk', 'portability': {}, 'locality': {}, 'subject': '
Twitter'}, 'time': 14.627016544342041, 'post': {'rewrite_acc': [1.0], 'locality': {}, 'portability': {}}}
05/13/2026 10:47:15 - INFO - easyeditor.editors.editor -   0 editing: The current CEO of Twitter is -> Linda Yaccarino
 {'pre': {'rewrite_acc': [0.6], 'portability': {}}, 'case_id': 0, 'requested_rewrite': {'prompt': 'The current CEO of Tw
itter is', 'target_new': 'Linda Yaccarino', 'ground_truth': 'Elon Musk', 'portability': {}, 'locality': {}, 'subject': '
Twitter'}, 'time': 14.627016544342041, 'post': {'rewrite_acc': [1.0], 'locality': {}, 'portability': {}}}
  ES: ✅ 直接问 -> The current CEO of Twitter is Linda Yaccarino, who was previ
  PS: ❌ 同义改写 -> Who is the CEO of Twitter?

TWTR, -0.35% The company's CEO i
  NS: ✅ 无关事实 -> The CEO of Apple is a man named Tim Cook. He's a very nice g
```

实验结果：

指标	成功数	总数	得分
ES（编辑成功率）	10	10	100%
PS（泛化性）	5	10	50%
NS（局部性）	6	10	60%